

Lukas Delaye

PhD student in astrophysics

Laboratoire d'Instrumentation et de Recherche en Astrophysique (LIRA) — Observatoire de Paris - PSL

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EDUCATION

- Master's degree (M2) in Astronomy and Astrophysics** 2024 - 2025
Université Paris-Saclay & Observatoire de Paris - PSL (Paris, France)
- Master's degree (M2) in higher education training in physics** 2023 - 2024
Obtaining agrégation in physics (recruitment of physics teacher)
Université Paris-Saclay & École normale supérieure Paris-Saclay (Orsay, France)
- Magistère in fundamental physics, Orsay** 2021 - 2024
Université Paris-Saclay (Orsay, France)
- Scientific preparatory classes for Grandes Écoles, PCSI / PC** 2019 - 2021
Lycée Camille Vernet (Valence, France)
- Scientific Baccalauréat, physics and chemistry major** 2016 - 2019
Lycée Polyvalent la Saulaie (Saint-Marcellin, France)

RESEARCH EXPERIENCES

- PhD in astrophysics, Observatoire de Paris - PSL (Meudon, France)** 2025 - 2028
- **Title of the thesis:** "Imaging exoplanets with the Roman Space Telescope".
 - **Topics:** *exoplanets, direct imaging, high-contrast imaging, adaptive optics, wavefront sensing and control, differential imaging, coronagraphy, numerical simulations, optical bench, Roman Space Telescope.*
 - **Supervisors:** ✉ Johan Mazoyer & ✉ Axel Potier.
 - **Professional investment:** Member of Roman Coronagraph Community Participation (CPP).
- M2 research internship, Observatoire de Paris - PSL (Meudon, France)** March 2025 - July 2025
- **Title of the internship:** "Coherence Differential Imaging (CDI) for the direct detection of Exoplanets".
 - **Topics:** *exoplanets, direct imaging, high-contrast imaging, adaptive optics, wavefront sensing and control, differential imaging, numerical simulations, VLT/SPHERE.*
 - **Supervisors:** ✉ Axel Potier & ✉ Raphaël Galicher.
- M2 Methodological project, Institut d'Astrophysique Spatiale (Orsay, France)** 2024 - 2025
- **Title of the project:** "Study of a photodissociation region".
 - **Topics:** *star formation, photodominated regions, radiative processes, data analysis, numerical simulation, JWST.*
 - **Supervisors:** ✉ Emilie Habart.
- L3 research internship, Institut d'Astrophysique Spatiale (Orsay, France)** June 2022 - July 2022
- **Title of the internship:** "Analysis of the aliphatic component of meteorites using infrared hyperspectral imaging".
 - **Topics:** *astrochemistry, infrared spectroscopy, hyperspectral analysis, Ryugu and the Hayabusa 2 mission.*

- **Supervisors:** ✉ Zélia Dionnet & ✉ Zahia Djouadi.

High school internship, *CERN - European Organization for Nuclear Research*

October 2017

- **Title of the internship:** "French High School Student Internship Programme".
- **Topics:** *Particle accelerator, magnetism, magnets.*

TEACHING

Contract teacher, *Observatoire de Paris - PSL (Paris, France)*

2025 - 2028

- Responsible for teaching 64 hours per year during the doctoral thesis.
- **Responsibilities:** *optics practicals in the first and second years of CPES, astronomical observation practicals in CPES, supervision week at the *Haute Provence Observatory (CNRS)* for the DU-ECU and for high school and college teachers, preparation for the DU-ECU cosmology TD, solar physics practicals for the DU-ECU, science festival: supervision of activities and visits to the domes of the Paris Observatory.*

Oral examiner, *Lycée Marcelin Berthelot (Saint-Maur-des-Fossés, France)*

2024 - 2025

- Two hours of oral practice per week in Scientific preparatory classes for Grandes Écoles. Preparation of second-year students (PC) for final recruitment exams.

Teaching intern, *Lycée Louis Blaringhem (Béthune, France)*

May 2023

- Teaching internship for a teacher in high school classes.
- **Supervisors:** ✉ Benoit Decke

Teaching intern, *Lycée Polyvalent la Saulaie (Saint-Marcellin, France)*

January 2022

- Teaching internship for a teacher in high school classes.
- **Supervisors:** ✉ Myriam Lutrin

SKILLS

Programming languages: Python, C, C++.

Teaching and research: L^AT_EX, Git environment (Github, Gitlab, GitKraken).

Languages: french (native), english (C1), spanish (B1).

WORKS

Peer-reviewed Publications

- DIONNET, Zélia, DJOUADI, Zahia, **DELAYE, Lukas**, et al. Methylene-to-methyl ratio variability in Ryugu samples: Clues to a heterogeneous aqueous alteration. *Meteoritics & Planetary Science*, **2025**, vol. 60, no 2, p. 273-285.

Conference Proceedings

- **DELAYE, Lukas**, LAGINJA, Iva, POTIER, Axel, et al. Enhanced wavefront sensing for the Roman Coronagraph Instrument: Gaussian probes and compact model validation. In: *SPIE Astronomical Telescopes + Instrumentation*. **in prep.**

Oral Presentations

- **DELAYE, Lukas**. The Roman Coronagraph Instrument: paving the technological path toward Earth-like exoplanet detection. *Elbereth*. Paris, France. **2026**

Conference Posters

- **DELAYE, Lukas**, POTIER, Axel, GALICHER, Raphael, et al. Simulated parametric study of coherent differential imaging for exoplanet detection with SPHERE. In: Spirit of Lyot 6. **2026**.
- **DELAYE, Lukas**, LAGINJA, Iva, POTIER, Axel, et al. Enhanced wavefront sensing for the Roman Coronagraph Instrument: Gaussian probes and compact model validation. In: SPIE Astronomical Telescopes + Instrumentation. **in prep.**

INTERESTS

Music: drummer in a student rock band, amateur guitarist, music creation using professional software.

Sports and Esports: triathlon (best result: participation in the semi-finals of the French Triathlon Championship), hiking, amateur video game competitions.

Other: photography, video, cinema